Southern Nuclear

David R. Vineyard
Vice President – Plant Hatch

Hatch Nuclear Plant
11028 Hatch Parkway North
Baxley, GA 31513
912 537 5859 tel
912 366 2077 fax

April 13, 2017

Docket Nos.: 50-321
50-366

NL-17-0637

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D. C. 20555-0001

Edwin I. Hatch Nuclear Plant Unit 1 & 2
Licensee Event Report 2017-003-00
Inoperable Secondary Containment Due to Incorrect Penetration Flange Installation

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(v)(C), Southern Nuclear Operating Company (SNC) hereby submits the enclosed Licensee Event Report.

This letter contains no NRC commitments. If you have any questions, please contact Greg Johnson at (912) 537-5874.

Respectfully submitted,

David R. Vineyard
Vice President – Hatch

DRV/jcb

Enclosure: LER 2017-003-00

cc: Southern Nuclear Operating Company
Mr. S. E. Kuczynski, Chairman, President & CEO
Mr. D. G. Bost, Executive Vice President & Chief Nuclear Officer
Mr. D. R. Vineyard, Vice President – Hatch
Mr. M. D. Meier, Vice President – Regulatory Affairs
Mr. R. D. Gayheart, Fleet Operations General Manager
Mr. B. J. Adams, Vice President – Engineering
Mr. G. L. Johnson, Regulatory Affairs Manager - Hatch
RTYPE: CHA02.004

U. S. Nuclear Regulatory Commission
Ms. C. Haney, Regional Administrator
Mr. R. Hall, NRR Project Manager – Hatch
Mr. D. H. Hardage, Senior Resident Inspector – Hatch

Edwin I. Hatch Nuclear Plant Unit 1 and 2

Licensee Event Report 2017-003-00

Inoperable Secondary Containment Due to Incorrect Penetration Flange Installation

NRC FORM 366 (06-2016)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 10/31/2018
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)		Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.	

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
Edwin I. Hatch Nuclear Plant Unit 1	05000321	1 OF 3

4. TITLE
Inoperable Secondary Containment Due to Incorrect Penetration Flange Installation

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME: Edwin I. Hatch Nuclear Plant Unit 2	DOCKET NUMBER: 05000366
02	17	2017	2017	- 003	- 00	4	13	2017	FACILITY NAME	DOCKET NUMBER

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
100	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER *(Include Area Code)*
Edwin I. Hatch / Jimmy Collins – Licensing Supervisor	912-537-2342

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX

14. SUPPLEMENTAL REPORT EXPECTED		15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)*	☒ NO				

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On February 17, 2017 at 1414 EST, with Unit 1 at 100 percent rated thermal power and Unit 2 at 0 percent rated thermal power due to a scheduled refueling outage, it was identified during an operations walkdown that a temporary blind flange was installed incorrectly, rendering secondary containment inoperable. In support of the Unit 2 Hardened Containment Vent System Wetwell Project, a temporary blind flange was to be installed downstream of the Standby Gas Treatment (SBGT) inlet isolation valve to prevent a breach of secondary containment. It was subsequently identified that upon removal of the SBGT isolation valve, the blind flange was installed on the wrong side of the piping. At 1503 EST on February 17, 2017, the blind flange was moved to the correct side of the piping and secondary containment was declared operable.

The cause of the event is due to having a lack of oversight, a lack of intrusiveness, and an inadequate challenge of the work instructions provided to the supplemental workers performing the work. Workers assumed that the work instructions meant for the blind flange to be installed downstream of primary containment rather than downstream of the SBGT inlet isolation valve. The work order planning procedure will be updated to require Southern Nuclear Operating Company (SNC) verification and signature for the installation of flanges and similar equipment.

NARRATIVE

Event Description

On February 17, 2017 at 1414 EST, with Unit 1 at 100 percent rated thermal power and Unit 2 at 0 percent rated thermal power due to a scheduled refueling outage, it was identified during an operations walkdown that a temporary blind flange was installed incorrectly, rendering secondary containment inoperable. The temporary blind flange was installed on February 8, 2017 at 0920 EST in support of the Unit 2 Hardened Containment Vent System (HCVS) Wetwell Project. As part of the project, the Unit 2 Standby Gas Treatment (SBGT) inlet isolation valve which separates the primary containment purge and inerting system from the SBGT system was removed from the line for local leak rate testing. During the period of time the isolation valve was to be removed, a temporary blind flange was to be installed on the downstream side of the opening to prevent a breach of secondary containment. After finding the temporary blind flange on the upstream side of the opening, it was revealed that upon removal of the SBGT isolation valve, the blind flange was installed on the wrong side of the opening. This opened a pathway outside of containment when containment modes changed on February 8, 2017 at 1035 EST when the Unit 2 reactor building was opened to the environment due to a scheduled refueling outage, resulting in a breach of secondary containment.

The as-found configuration rendered secondary containment inoperable for Unit 1. Actions were made to immediately suspend fuel movement and restore secondary containment by putting the flange on the correct side of the opening. At 1503 EST on February 17, 2017, the blind flange was moved to the correct side of the flange and secondary containment was declared operable.

Event Cause Analysis

The cause of the event is due to having a lack of oversight, a lack of intrusiveness, and an inadequate challenge of the work instructions provided to the supplemental workers performing the work. The supplemental maintenance workers had an incorrect mindset and assumed that the blind flange was to be installed on the primary containment side of the containment piping to provide protection for the reactor. Therefore, the work instructions to install the blind flange on the "downstream" side of the opening were not followed. Workers assumed that the work instructions meant for the blind flange to be installed downstream of primary containment rather than downstream of the SBGT inlet isolation valve. The associated work order package did not contain annotated drawings showing the exact installation location of the blind flange. Workers consequently proceeded with the installation of the blind flange on the incorrect side of the opening without having their assumptions verified.

Safety Assessment

This event is reportable per 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by Technical Specifications and also per 10 CFR 50.73(a)(2)(v)(C) as a condition that at the time of discovery, could have prevented the fulfillment of the safety function of a system needed to control the release of radioactive material.

The secondary containment system contains the primary containment system and other nuclear systems, and limits the ground-level release of airborne radioactive material. It provides a means for a controlled elevated release of the building atmosphere such that offsite doses from a fuel-handling or loss-of-coolant accident (LOCA) will be below the guideline values stated in 10 CFR 50.67. It is designed to provide secondary containment when the primary containment is closed and in service, and to provide primary containment when the primary containment is open, e.g., during a refueling outage. Secondary containment is isolated on the same signals that actuate the Standby Gas Treatment (SBGT) system. Activation of the SBGT demonstrates the integrity of the reactor building (secondary containment). The SBGT establishes and maintains a negative pressure (0.2 inches water) within the secondary containment.

The following four types of accidents are analyzed for potential dose consequences: a LOCA, a main steam line break (MSLB), a control rod drop accident (CRDA), and a fuel handling accident (FHA). Of these four accidents, only the LOCA analysis relies upon operability of secondary containment and the SBGT system to mitigate doses. This analysis assumes secondary containment draws down to a differential pressure (dp) of 0.2 inches water gauge with respect to the atmosphere in 10 minutes. Therefore, a secondary containment draw down to greater than or equal 0.2 inches in less than 10 minutes would be conservative and is bounded by this analysis.

Based on this information above, there were multiple instances that demonstrated the secondary containment and SBGT system would have been able to perform its safety function in the event of an accident, even with the blind flange being installed incorrectly. On February 16, 2017 at 1319 EST, due to maintenance activities, the SBGT system auto started and secondary containment isolated. Secondary containment was drawn down to greater than 0.2 inches water gauge in 30 seconds. Also on February 17, 2017 at 1021 EST, due to maintenance activities, the SBGT system auto started and secondary containment isolated. Secondary containment was drawn down to greater than 0.2 inches water gauge in 50 seconds. Therefore, although the lack of a qualified isolation device to limit the release of radioactive material constitutes a loss of secondary containment integrity, it can be concluded that secondary containment would have been able to perform its safety function. This event is considered to have very low safety significance.

Corrective Actions

Stand downs were held with site and supplemental personnel to discuss the event and contributing factors. All operations and maintenance supervision also met with the Site Plant Manager and/or the Site Vice President to discuss the event and learnings. The work order planning procedure will be updated to require Southern Nuclear Operating Company (SNC) verification and signature of work performed by supplemental workers for the installation of flanges and similar equipment utilized in maintaining operability of Technical Specification related equipment.

Previous Similar Events

None.